

◆ A.9 — Delay Stability (Lyapunov–Krasovskii, Final Version)

A.9.1 Delayed System

$$\dot{\Psi}(t) = F(\Psi(t)) - L\Psi(t-\tau), \quad \tau \in [0, \tau_{\max}]$$

Assumptions:

- F is globally Lipschitz
- Dissipativity:

$$\langle \Psi, F(\Psi) \rangle \leq -c\|\Psi\|^2, \quad c > 0$$

- $L \succeq 0$, with:

$$\|L\| = \lambda_{\max}(L)$$

A.9.2 Lyapunov–Krasovskii Functional

$$V(t) = \|\Psi(t)\|^2 + \int_{t-\tau}^t \|\Psi(s)\|^2 ds$$

Bounds:

$$\|\Psi(t)\|^2 \leq V(t) \leq (1+\tau) \sup_{s \in [t-\tau, t]} \|\Psi(s)\|^2$$

A.9.3 Time Derivative

$$\begin{aligned} \dot{V}(t) &= \\ 2\langle \Psi(t), \dot{\Psi}(t) \rangle &+ \|\Psi(t)\|^2 - \|\Psi(t-\tau)\|^2 \end{aligned}$$

Substituting dynamics:

$$\begin{aligned} \dot{V} &= \\ 2\langle \Psi, F(\Psi) \rangle &- 2\langle \Psi, L\Psi(t-\tau) \rangle \\ &+ \|\Psi(t)\|^2 - \|\Psi(t-\tau)\|^2 \end{aligned}$$

A.9.4 Bounding Terms

Dissipativity:

$$2\langle \Psi, F(\Psi) \rangle \leq -2c\|\Psi\|^2$$

Coupling term (Young's inequality):

For any $\epsilon > 0$:

$$\begin{aligned} 2|\langle \Psi, L\Psi_{-\tau} \rangle| &\leq \\ \|L\| \left(\|\Psi\| \|\Psi_{-\tau}\| \right) &\leq \epsilon \|\Psi\|^2 + \frac{\|L\|^2}{\epsilon} \|\Psi_{-\tau}\|^2 \end{aligned}$$

$$\left(\frac{1}{\epsilon} \left(\|\Psi\|^2 + \|\Psi_{\tau}\|^2 \right) \right)$$

where $\Psi_{\tau} = \Psi(t - \tau)$.

A.9.5 Differential Inequality

$$\begin{aligned} \dot{V}(t) &\leq \\ & - \left(2c - \frac{1}{\epsilon} \right) \|\Psi(t)\|^2 \\ & - \\ & \left(1 - \frac{\|L\|}{\epsilon} \right) \|\Psi(t - \tau)\|^2 \end{aligned}$$

Choose ϵ such that:

$$\|L\| < \epsilon < \frac{2c-1}{\|L\|}$$

Then:

$$\dot{V}(t) \leq -\alpha V(t), \quad \alpha > 0$$

A.9.6 Stability Condition (Delay Margin)

A sufficient condition for exponential stability is:

$$2c - 1 > \|L\|$$

For delay-dependent stability:

$$\boxed{\tau_{\max} < \frac{\alpha}{\beta}}$$

for suitable constants $\alpha, \beta > 0$.

A.9.7 Result

$$\begin{aligned} \dot{V}(t) &\leq -\alpha V(t) \\ &\Rightarrow V(t) \leq V(0)e^{-\alpha t} \\ &\Rightarrow \Psi(t) \rightarrow \Psi^* \end{aligned}$$

FINAL RESULT — A.9

$$\boxed{\text{The system remains exponentially stable under bounded time delay.}}$$

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